Maximum Weight Perfect Matching on the Aztec Diamond

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Illinois Geometry Lab
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The Aztec Diamond D_n

D_n consists of *lattice squares* with centers (x, y) satisfying $|x| + |y| \leq n$.

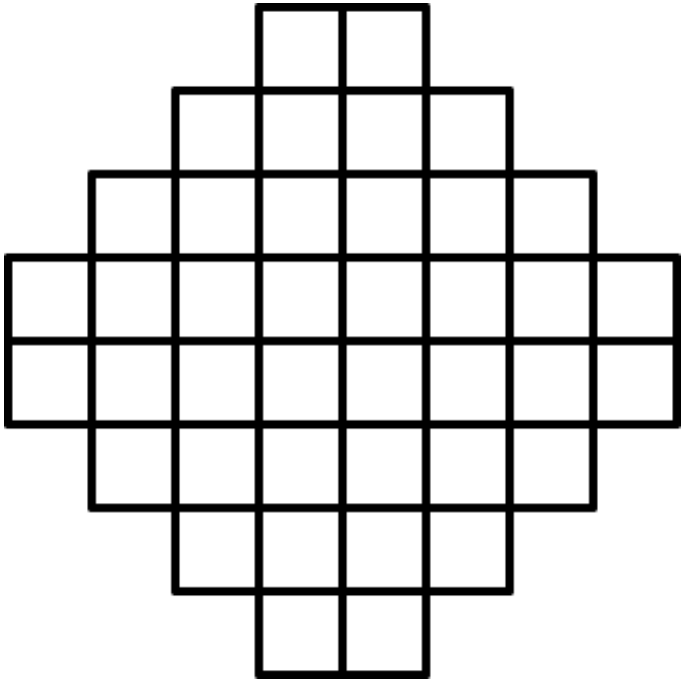
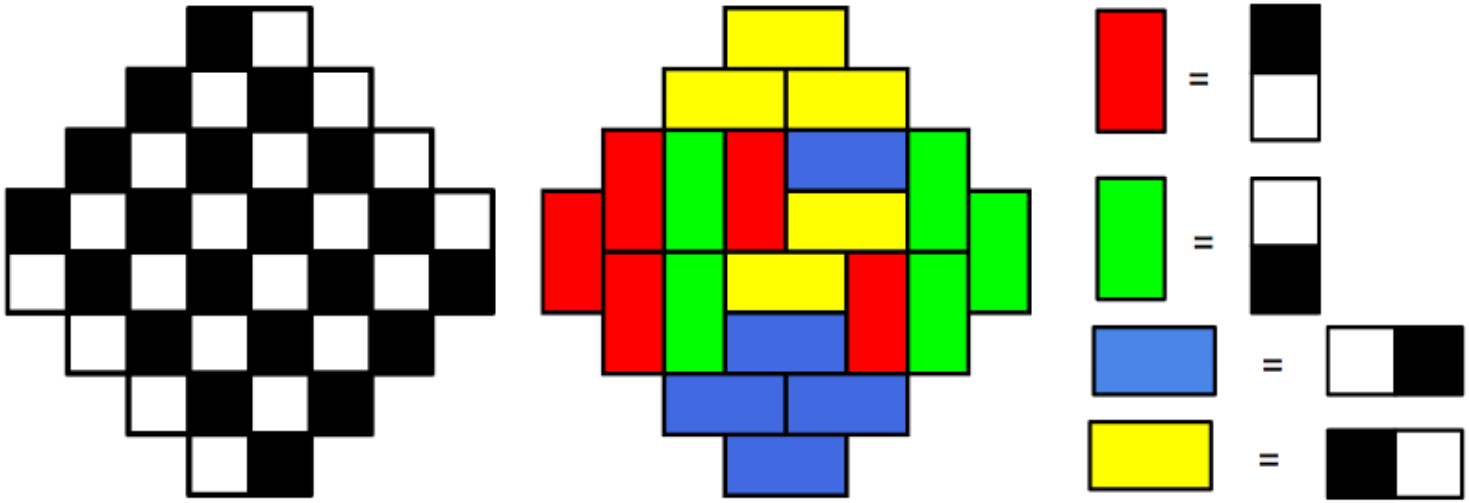


Figure: D_4

Credit: `https://en.wikipedia.org/wiki/Aztec_diamond`

Domino Tiliing

A domino tiling is a packing of 2x1 rectangles into the Aztec Diamond.



Credit: `https://en.wikipedia.org/wiki/Aztec_diamond`

Perfect Matching

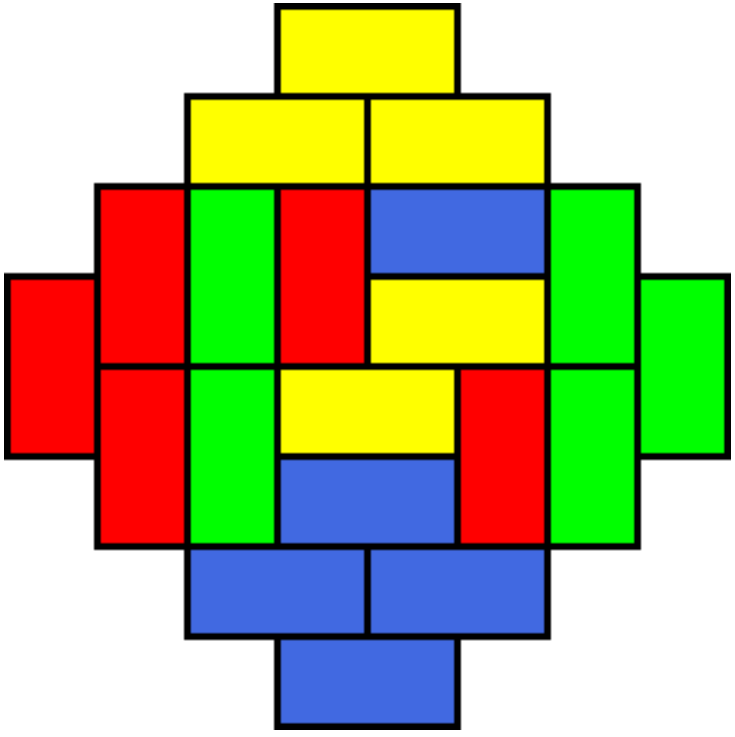


Figure: Domino tilling

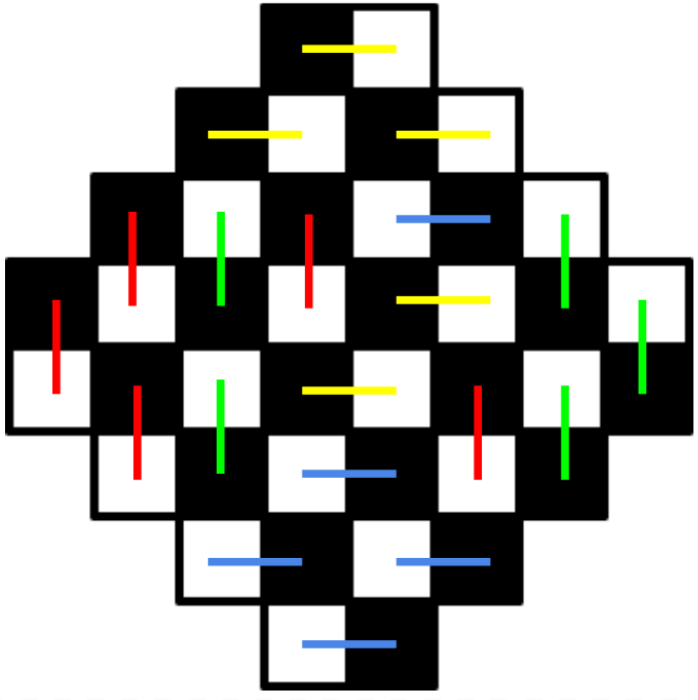


Figure: Perfect matching

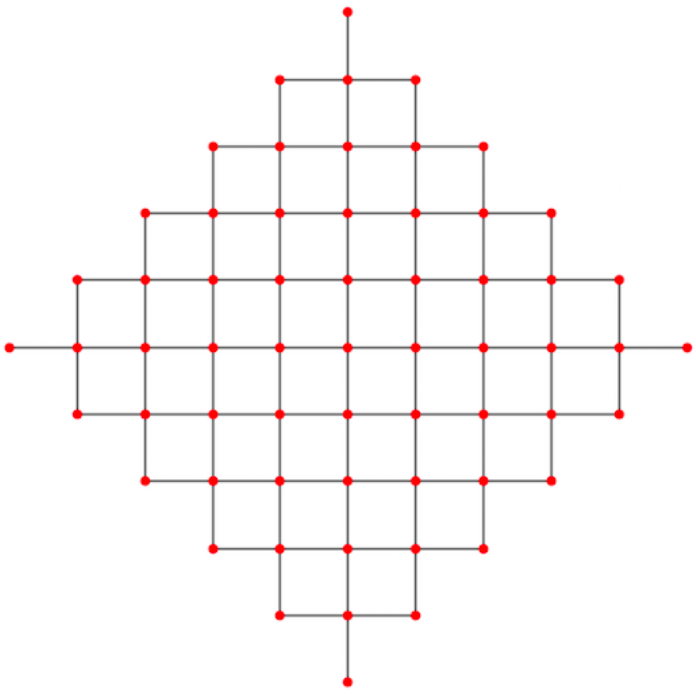
Project Goals

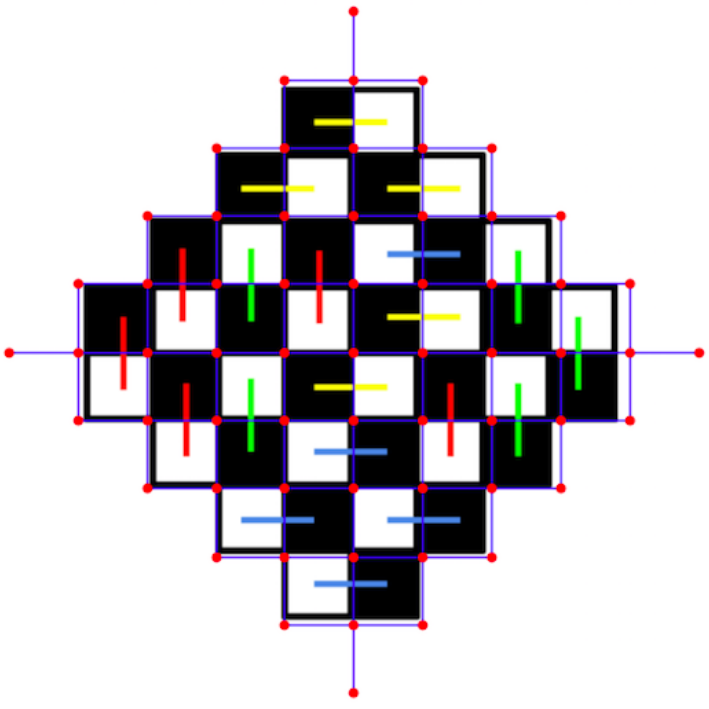
- Previous results (Arctic Circle Theorem): Analyze a randomly chosen perfect matching on the Aztec diamond.

- What we do: Given random weights to the edges in the Aztec diamond, analyze the maximum weight perfect matching.

Height Function

Given a perfect matching, how to define the height function?





Sample Output

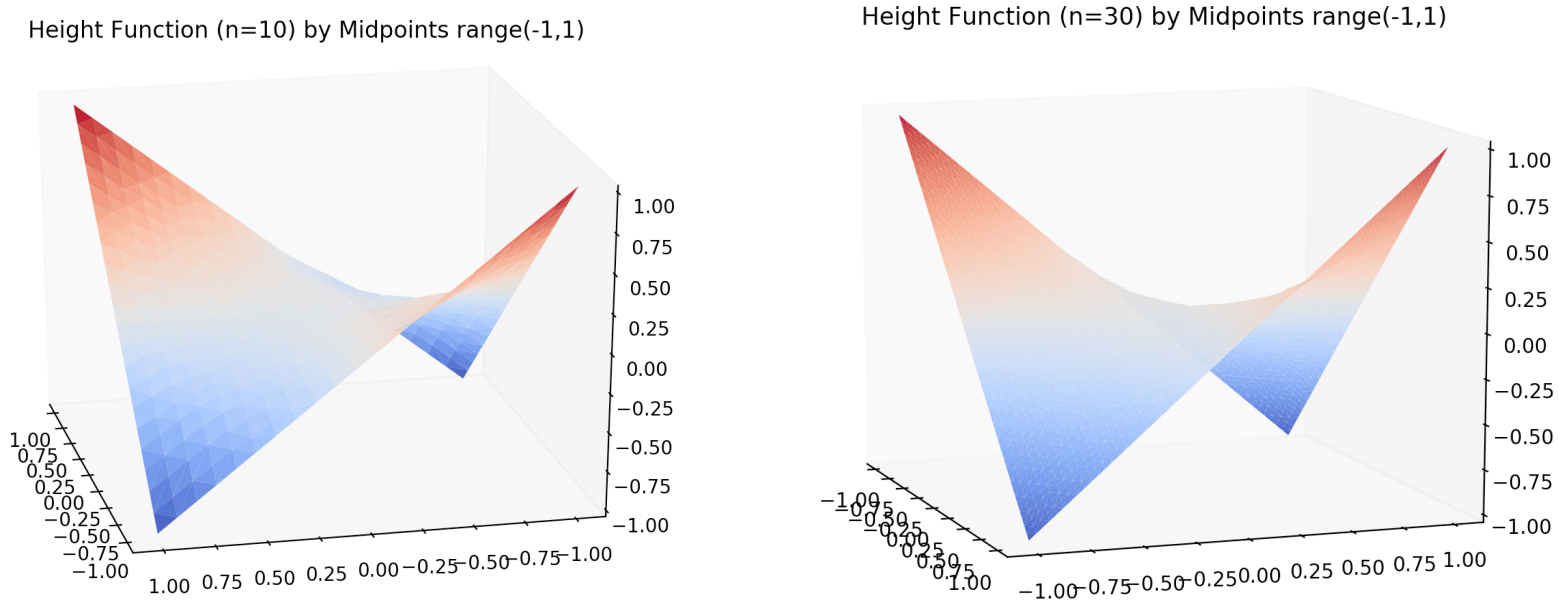


Figure: Sample surface for $n = 10$ and $n = 30$ with random weights.

Conjecture

Conjecture: This centered and scaled surface is converging to $z = xy$ surface.

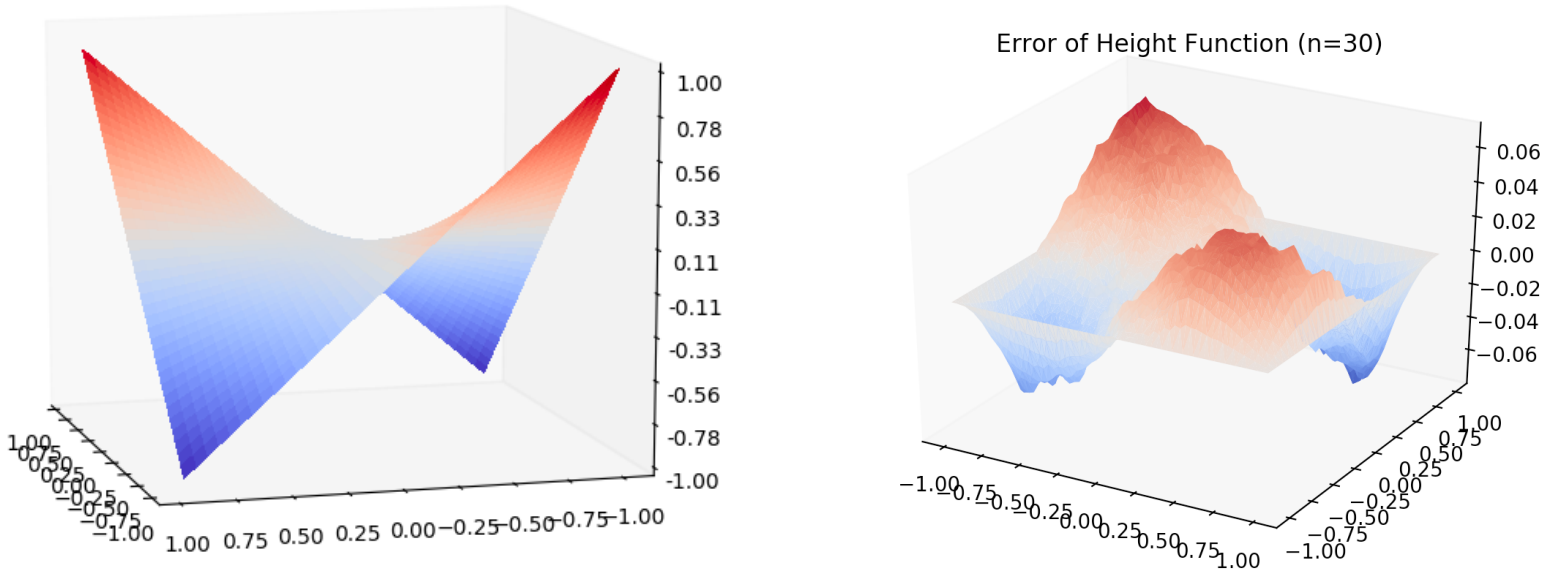


Figure: $z = xy$ surface and error in a sample.

Future work

- Asymptotic distribution of error.
- Behaviors on other lattices, such as triangular lattice.
- Optimizing algorithm — currently too slow.

Thank you!